

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theor y	Pract .	Tut.	Total
2113115	Analysis of Algorithm Lab	--	2	-	--	2	-	2

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test1	Test 2	Avg. of 2 Tests				
2113115	Analysis of Algorithm Lab	--	--	--	--	25	25	50

Lab Objectives:

1. To introduce the methods of designing and analyzing algorithms
2. Design and implement efficient algorithms for a specified application
3. Strengthen the ability to identify and apply the suitable algorithm for the given real-world problem.
4. Analyze worst-case running time of algorithms and understand fundamental algorithmic problems

Lab Outcomes: At the end of the course, the students will be able to

1. Implement the algorithms using different approaches.
2. Analyze the complexities of various algorithms.
3. Compare the complexity of the algorithms for specific problem.

Prerequisite: Basic knowledge of programming and data structure

DETAILED SYLLABUS:

Sr. No.	Module	Detailed Content	Hours	LO Mapping
0	Prerequisite	Basic knowledge of programming and data structure		
I	Introduction	Performance analysis- Master Method, space, and time complexity Growth of function, Big-Oh, Omega Theta notation Mathematical background for algorithm analysis. Analysis of selection sort, insertion sort.	2	LO1, LO2, LO3
II	Divide and Conquer Approach	General method, Merge sort, Quick sort, Analysis of Binary search. Self-learning Topics: Finding minimum and maximum algorithms and their Analysis, Strassen's Algorithm	2	LO1, LO2, LO3
III	Greedy Method Approach	General Method, Single source shortest path: Dijkstra Algorithm Fractional Knapsack problem, Minimum cost spanning trees: Kruskal and Prim's algorithms Self-learning Topics: Job sequencing with deadlines	2	LO1, LO2, LO3
IV	Dynamic Programming	General Method, Multistage graphs	2	LO1, LO2,

	Approach	All pair shortest path: Floyd Warshall Algorithm, 0/1 knapsack Problem, Travelling Salesperson problem, Longest common subsequence Self-learning Topics: Bellman Ford Algorithm		LO3
V	Backtracking and Branch and bound	General Method, Backtracking: N-queen problem, Sum of subsets, Graph coloring Branch and Bound: Travelling Salesperson Problem, 15 Puzzle problem	2	LO1, LO2, LO3
VI	String Matching Algorithms	The Naïve string-matching algorithm, The Rabin Karp algorithm, The Knuth-Morris-Pratt algorithm	2	LO1, LO2, LO3

Text Books:

1. T. H. Cormen, C.E. Leiserson, R. L. Rivest, and C. Stein, “Introduction to algorithms”, 2nd Edition, PHI Publication 2005.
2. Ellis Horowitz, Sartaj Sahni, S. Rajsekar. “Fundamentals of computer algorithms” University Press.

Reference Books:

1. Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, “Algorithms”, Tata McGraw- Hill Edition.
2. S. K. Basu, “Design Methods and Analysis of Algorithm”, PHI

Online Resources:

Sr. No.	Website Name
1.	https://nptel.ac.in/courses/106/106/106106131/
2.	https://swayam.gov.in/nd1_noc19_cs47/preview
3.	https://www.coursera.org/specializations/algorithms
4.	https://www.mooc-list.com/tags/algorithms

Sr No	List of Assignments / Tutorials	Hrs
01	Assignment covers the topics from first three units limited to three Questions	2
02	Assignment covers the topics from Last three units limited to three Questions	2

Suggested list of Experiments.

Sr No	Title of Experiments	Hrs
01	Experiment based on common mathematical functions.(Selection sort, Insertion sort)	2
02	Experiment based on divide and conquers approach. (Merge sort, Quick sort, Binary search)	2
03	Experiment based on greedy approach.(Single source shortest path- Dijkstra Fractional Knapsack problem, Minimum cost spanning trees-Kruskal and Prim’s algorithm)	2
04	Experiment using dynamic programming approach (All pair shortest path- Floyd Warshall, 0/1 knapsack)	2
05	Travelling salesperson problem Longest common subsequence	2
06	Experiment based on graph Algorithms (BFS, DFS , etc)	2
07	Experiment using Backtracking strategy. (N-queen problem, Sum of subsets, Graph coloring)	2
08	Experiment using branch and bound strategy.	2
09	Experiment based on string matching/amortized analysis (The Naïve string-matching Algorithms , The Rabin Karp algorithm, The Knuth-Morris-Pratt algorithm.	2
10	Implementation Min-Max Algorithm	2
11	Implementation of Job Sequencing with deadlines.	2
12	Implementation of Bellman Ford Algorithm using Dynamic programming	2

Assessment:

Term Work: Term Work shall consist of at least 10 to 12 practicals' based on the above list. Also, Term work Journal must include at least 2 assignments.

Term Work Marks: 25 Marks (Total marks) = 15 Marks (Experiment) + 5 Marks (Assignments) + 5 Marks (Attendance)

Practical& Oral Exam: An Oral & Practical exam will be held based on the above syllabus.